C-F-202F
December 31, 1985
SUPERSEDING
C-F-202E
March 4, 1970

FEDERAL SPECIFICATION

# FELT SHEET (HAIR) AND FELT ROLL (HAIR)

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cattle hair felt as manufactured by the felted and fulled process in sheet and roll form.

1.2 Classification.

1.2.1 Types and classes. The felt shall be of the following types and classes, as specified (see 6.2 and 6.5). The thickness and weight designations shall be as required in table I, as specified (see 6.2):

Type H-26-S
Class U - Untreated
Class N - Neutralized

Type H-26-R
Class U - Untreated
Class N - Neutralized

Type H-34-S
Class U - Untreated
Class N - Neutralized

Type H-34-R
Class U - Untreated
Class N - Neutralized

Type H-43-S
Class U - Untreated
Class N - Neutralized

AMSC N/A

FSC 8305

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

Type H-43-R
Class U - Untreated
Class N - Neutralized

Type H-56-S
Class U - Untreated
Class N - Neutralized

Type H-56-R
Class U - Untreated
Class N - Neutralized

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications

| | |
|---|---|
| T-T-871 | - Twine, Cotton, Wrapping |
| T-T-911 | - Twine, Fibrous, Jute |
| UU-P-268 | - Paper, Kraft, Wrapping |
| CCC-C-429 | - Cloth, Osnaburg, Cotton |
| CCC-C-467 | - Cloth, Burlap, Jute (or Kenaf) |
| PPP-B-591 | - Boxes, Fiberboard, Wood-Cleated |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner |
| PPP-B-636 | - Boxes, Shipping, Fiberboard |
| PPP-T-45 | - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing |

Federal Standards

| | |
|---|---|
| FED-STD-123 | - Marking for Shipment (Civil Agencies) |
| FED-STD-191 | - Textile Test Methods |

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

(Single copies of this document and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

| | |
|---|---|
| MIL-F-2312 | - Felt, Hair or Wool: Mildew Resistant and Moisture Resistant Treatment For |
| MIL-L-10547 | - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible |
| MIL-T-40625 | - Tubing, Bias Sewn (Burlap or Osnaburg), Cloth |

Military Standards

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | - Marking for Shipment and Storage |

Laws and Regulations

Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580.)

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials (ASTM):

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.2, 6.2, and 6.3).

3.2 Material.

3.2.1 Fiber. The felt shall be made predominately from washed cattle hair, which shall be the main fiber constituent of the felt, but not to the exclusion of other animal hair. The fiber content of all felt shall be not less than 95 percent hair fibers. These requirements shall be tested as specified in 4.4.

3.2.2 Extractable matter. Chloroform and water extractable matter shall not exceed 16.0 percent by weight, based on the bone-dry weight of the test specimen. These requirements shall be tested as specified in 4.4.

3.2.3 Weight and thickness. The felt shall conform to the requirements for thickness in inches and weight in pounds per square yard, as specified in table I, when tested as specified in 4.4.

TABLE I. Thickness and weight of finished hair felt

| Types (classes U and N) | Thickness of finished felt (inches) | | | Weight of finished felt (pounds/square yard) | | |
|---|---|---|---|---|---|---|
| | Nominal | Min. | Max. | Nominal | Min. | Max. |
| H-26-S | 0.188 | 0.170 | 0.206 | 2.26 | 2.03 | 2.49 |
| and | 0.250 | 0.225 | 0.275 | 3.00 | 2.70 | 3.30 |
| H-26-R | 0.313 | 0.281 | 0.345 | 3.75 | 3.37 | 4.13 |
| | 0.375 | 0.338 | 0.412 | 4.50 | 4.05 | 4.95 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

TABLE I. Thickness and weight of finished hair felt - Continued

| Types (classes U and N) | Thickness of finished felt (inches) | | | Weight of finished felt (pounds/square yard) | | |
|---|---|---|---|---|---|---|
| | Nominal | Min. | Max. | Nominal | Min. | Max. |
| H-26-S | 0.500 | 0.450 | 0.550 | 6.00 | 5.40 | 6.60 |
| and | 0.625 | 0.562 | 0.688 | 7.50 | 6.75 | 8.25 |
| H-26-R | 0.750 | 0.675 | 0.825 | 9.00 | 8.10 | 9.90 |
| | 0.875 | 0.787 | 0.963 | 10.50 | 9.45 | 11.55 |
| | 1.000 | 0.900 | 1.100 | 12.00 | 10.80 | 13.20 |
| | 1.250 | 1.130 | 1.370 | 15.00 | 13.00 | 17.00 |
| | 1.500 | 1.350 | 1.650 | 18.00 | 16.00 | 20.00 |
| | 1.750 | 1.600 | 1.900 | 21.00 | 19.00 | 23.00 |
| | 2.000 | 1.850 | 2.150 | 24.00 | 22.00 | 26.00 |
| H-34-S | 0.125 | 0.112 | 0.137 | 2.00 | 1.80 | 2.20 |
| and | 0.188 | 0.170 | 0.206 | 3.00 | 2.70 | 3.30 |
| H-34-R | 0.250 | 0.225 | 0.275 | 4.00 | 3.60 | 4.40 |
| | 0.313 | 0.281 | 0.345 | 5.00 | 4.50 | 5.50 |
| | 0.375 | 0.338 | 0.412 | 6.00 | 5.40 | 6.60 |
| | 0.500 | 0.450 | 0.550 | 8.00 | 7.20 | 8.80 |
| | 0.625 | 0.562 | 0.688 | 10.00 | 9.00 | 11.00 |
| | 0.750 | 0.675 | 0.825 | 12.00 | 10.80 | 13.20 |
| | 0.875 | 0.787 | 0.963 | 14.00 | 12.60 | 15.40 |
| | 1.000 | 0.900 | 1.100 | 16.00 | 14.40 | 17.60 |
| | 1.250 | 1.130 | 1.370 | 20.00 | 18.00 | 22.00 |
| | 1.500 | 1.350 | 1.650 | 24.00 | 22.00 | 26.00 |
| | 1.750 | 1.600 | 1.900 | 28.00 | 26.00 | 30.00 |
| | 2.000 | 1.850 | 2.150 | 32.00 | 30.00 | 34.00 |
| H-43-S | 0.094 | 0.082 | 0.106 | 1.88 | 1.69 | 2.07 |
| and | 0.125 | 0.112 | 0.137 | 2.50 | 2.25 | 2.75 |
| H-43-R | 0.188 | 0.170 | 0.206 | 3.75 | 3.37 | 4.13 |
| | 0.250 | 0.225 | 0.275 | 5.00 | 4.50 | 5.50 |
| | 0.313 | 0.281 | 0.345 | 6.25 | 5.62 | 6.88 |
| | 0.375 | 0.338 | 0.412 | 7.50 | 6.75 | 8.25 |
| | 0.500 | 0.450 | 0.550 | 10.00 | 9.00 | 11.00 |
| H-43-S | 0.625 | 0.562 | 0.688 | 12.50 | 11.25 | 13.75 |
| and | 0.750 | 0.675 | 0.825 | 15.00 | 13.50 | 16.50 |
| H-43-R | 0.875 | 0.787 | 0.963 | 17.50 | 15.75 | 19.25 |
| | 1.000 | 0.900 | 1.100 | 20.00 | 18.00 | 22.00 |
| | 1.250 | 1.130 | 1.370 | 25.00 | 23.00 | 27.00 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

TABLE I. Thickness and weight of finished hair felt - Continued

| Types (classes U and N) | Thickness of finished felt (inches) | | | Weight of finished felt (pounds/square yard) | | |
|---|---|---|---|---|---|---|
| | Nominal | Min. | Max. | Nominal | Min. | Max. |
| H-43-S and H-43-R | 1.500 | 1.350 | 1.650 | 30.00 | 28.00 | 32.00 |
| | 1.750 | 1.600 | 1.900 | 35.00 | 33.00 | 37.00 |
| | 2.000 | 1.850 | 2.150 | 40.00 | 38.00 | 42.00 |
| H-56-S and H-56-R | 0.094 | 0.082 | 0.106 | 2.44 | 2.20 | 2.68 |
| | 0.125 | 0.112 | 0.137 | 3.25 | 2.92 | 3.58 |
| | 0.188 | 0.170 | 0.206 | 4.88 | 4.39 | 5.37 |
| | 0.250 | 0.225 | 0.275 | 6.50 | 5.80 | 7.15 |
| | 0.313 | 0.281 | 0.345 | 8.14 | 7.33 | 8.95 |
| | 0.375 | 0.338 | 0.412 | 9.75 | 8.77 | 10.73 |
| | 0.500 | 0.450 | 0.550 | 13.00 | 11.70 | 14.30 |
| | 0.625 | 0.562 | 0.688 | 16.25 | 14.62 | 17.88 |
| | 0.750 | 0.675 | 0.825 | 19.50 | 17.55 | 21.45 |
| | 0.875 | 0.787 | 0.963 | 22.75 | 20.75 | 24.75 |
| | 1.000 | 0.900 | 1.100 | 26.00 | 24.00 | 28.00 |
| | 1.250 | 1.130 | 1.370 | 32.50 | 30.50 | 34.50 |
| | 1.500 | 1.350 | 1.650 | 39.00 | 37.00 | 41.00 |
| | 1.750 | 1.600 | 1.900 | 45.50 | 43.50 | 47.50 |
| | 2.000 | 1.850 | 2.150 | 52.00 | 50.00 | 54.00 |

3.2.4 Roll length and width. Unless otherwise specified (see 6.2), the felt shall be furnished in rolls 48 inches ± 1.0 inch in width and in continuous length ranges in accordance with the following thickness schedule:

| Thickness (inches) | Length (yards) |
|---|---|
| 5/16 and under | 8-17 |
| Over 5/16 and less than 1/2 | 5-12 |
| 1/2 to 1 inclusive | 3-7 |
| 1 1/4 to 2 | 1-3 |



Source: https://assist.dla.mil -- Downloaded: 202640-26T11:04Z
Check the source to verify that this is the currentversion before use.

3.2.5 Sheet length and width. Unless otherwise specified (see 6.2), the felt shall be furnished in sheet form with trimmed edges in the following sizes as specified:

| Sheet size, inches ( ± 1/2 inch in either direction) | | | |
|---|---|---|---|
| 30 x 24 | 30 x 48 | 36 x 36 | |
| 30 x 36 | 36 x 24 | 36 x 48 | 36 x 60 |

3.2.6 Color. Unless otherwise specified (see 6.2), the felt shall be natural; the color resulting from the various shades of brown hair used shall be acceptable.

3.2.7 Organic acidity. When specified (see 6.2 and 6.4), all class "U" felt shall not show an organic acidity greater than 0.2 percent when tested as specified in 4.4.

3.2.8 Alkalinity and mineral acidity. When specified (see 6.2 and 6.4), all class "N" felts shall have a pH of 7.0 ± 1.0 when tested as specified in 4.4.

3.2.9 Mildew treatment. When specified (see 6.2), the felt shall be mildew-resistant treated in accordance with type I of MIL-F-2312. The physical and chemical characteristics of the treated felt shall conform to the requirements specified therein. Organic acidity, alkalinity, and mineral acidity requirements are not applicable to mildew-resistant treated felts.

3.3 Construction. The construction of all types and classes of the felt covered by this specification shall be that of a textile composed wholly of hair fibers physically interlocked by the inherent felting properties of animal hair fibers and produced by a suitable combination of mechanical work, chemical action, moisture, and heat, but without needle loom weaving, knitting, stitching, thermal bonding, or adhesives.

3.3.1 Flexibility. All types and classes of felt covered by this specification shall show no visible sign of cracking when tested as specified in 4.4.

3.3.2 Splitting resistance. All types and classes of felt covered by this specification shall have a minimum splitting resistance of 4.0 pounds in each direction when tested as specified in 4.4. This test is not applicable to felt of 1/8 inch and under in thickness. The split sections shall be visually inspected to ensure that needle looming has not been used during manufacture.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

3.3.3 Bulk density. The bulk density shall be as follows when tested as specified in 4.4:

Types H-26-S and R = 0.26
Types H-34-S and R = 0.34
Types H-43-S and R = 0.43
Types H-56-S and R = 0.56

3.4 Identification. Each roll and sheet shall be identified by the following methods as applicable.

3.4.1 Identification. Each roll shall have a piece ticket (see 6.2) (identification tag) attached to the edge at the inner end of the felt with not finer than 5-ply cotton string doubled to not less than 8 inches long or a plastic tag hanger of 5-inch minimum length. The piece ticket shall be made of not less than 15-point paper stock, and the finished surface shall not be calendered or pigmented to the degree that would cause the obliteration of printed, stamped, or typed markings. The piece ticket shall have a reinforced eyelet for attaching the typing cord or hanger and shall be legibly printed with water insoluble ink with the markings shown on figure 1. Entries on piece tickets (tags) shall be printed, stamped, or typed. The entries shall be complete and responsive with respect to the information required by the piece ticket legend (see figure 1). Hand lettered entries are permitted, but hand written entries are not permitted.

3.4.2 Labels. Each sheet of felt shall have a label attached to the upper right corner by staple or pin-ticket attachment. The label shall be made of not less than 15-point paper stock, and the color shall be white or light in intensity to permit easy reading of printed markings. The label shall be legibly printed with water insoluble ink with the information specified in 3.4.1, except that gross and net yards are replaced by width and length.

3.4.3 Fiber identification label. Felt manufactured under this specification shall be labeled in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.5 Workmanship. The finished felt shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check-test such items to determine the validity of the certification.

4.2 First article inspection. Before production is commenced, one square yard of roll felt or one sheet of felt, as applicable, of the finished felt shall be submitted or made ready for the contracting officer or his authorized representative for visual inspection of appearance, color, and finish. The sample shall be tested for physical and chemical properties as specified in 4.4. The approval of the preproduction sample authorizes the commencement of production but does not relieve the contractor of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production items.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.3.2 Examination of the end item. Defects found during the visual examination shall be classified in accordance with 4.3.2.1 through 4.3.2.3.2.

4.3.2.1 Visual examination. The felt shall be examined on one side only. However, for each successive roll or sheet in the sample, that side shall be alternated. Defects found during this examination shall be classified in accordance with table II. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. In rolled felt, a defect extending over more than 1 continuous linear yard shall be counted as one defect for each yard or fraction thereof in which it occurs. The lot size shall be expressed in units of 1 linear yard when rolls are specified and one sheet when sheets are specified. The sample unit shall be 1 linear yard when rolls are specified and one sheet when sheets are specified. The inspection level shall be I. The acceptable quality



Source: https://assist.dla.mil -- Downloaded: 2026-04-26T11:04Z
Check the source to verify that this is the current version before use.

levels (AQLs) shall be 0.65 for major defects and 4.0 for total defects, for a roll width of 4 feet and for sheets. The number of rolls from which the sample is to be taken shall be in accordance with table III. An approximate equal number of yards shall be examined from each roll.

TABLE II. Classification of visual defects

| Defects | Classification | |
|---|---|---|
| | Major | Minor |
| Cut, hole or tear - any size | X | |
| Thin or weak place - any size | X | |
| Felting rub - clearly noticeable but not resulting in a thin or weak place | | X |
| High place or lump | | X |
| Hardener rib | | X |
| Width (rolls only): | | |
| - more than 1/2 inch over the specified width | | X |
| - more than 1/2 inch less than the specified width | X | |
| Width and length (sheets only): | | |
| - more than 1/2 inch over the specified width or length | | X |
| - more than 1/2 inch less than the specified width or length | X | |
| Identification (ticket or label): | | |
| - missing, incorrect, incomplete, illegible, or insecurely attached | | X |

4.3.2.2 Examination for overall defects. The felt shall be examined for the overall defects listed below. Each defect shall be counted not more than once in a roll or sheet. The sample unit shall be one roll or sheet. The number of rolls or sheets selected shall be in accordance with table III. The lot shall be unacceptable if the defects found exceed the maximum number permitted in table III.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

Overall defects

Burrs, shives, or other foreign matter, any protrusion or accumulation visible from 3 feet
Uneven felting
Color not natural
Wrong type, roll, or sheet
Not clean

TABLE III. Examination criteria

| Lot size in yards or sheets | Sample size in Rolls 1/ | Sample size in Sheets 1/ | Maximum number of defects acceptable in sample Rolls | Maximum number of defects acceptable in sample Sheets |
|---|---|---|---|---|
| Up to 1200 | 3 | 7 | 0 | 0 |
| 1201 up to and including 3200 | 5 | 10 | 0 | 0 |
| 3201 up to and including 10,000 | 7 | 15 | 0 | 0 |
| 10,001 up to and including 35,000 | 10 | 25 | 0 | 1 |
| 35,001 up to and including 150,000 | 15 | 35 | 1 | 2 |
| 150,001 and over | 25 | 50 | 1 | 3 |

1/ If lot contains fewer than the required number of rolls or sheets, each roll or sheet in the lot shall be examined.

4.3.2.3 Length examination (rolls only).

4.3.2.3.1 Individual rolls. Each roll shall be examined for gross length. Any gross length found to be less than the minimum length specified or more than 1 yard less than the gross length marked on the roll ticket, or not in a continuous length, shall be considered a defect with respect to length. The



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

sample unit shall be one roll. The rolls used for the visual examination shall be used for this examination. The lot shall be unacceptable if one or more defects are found in the sample.

4.3.2.3.2 Examination for total yardage (rolls only). The lot shall be unacceptable if the total of the actual gross lengths in the sample is less than the total of the gross lengths marked on the roll tickets. The rolls used for the visual examination shall be used for this examination.

4.3.3 Packaging inspection. An examination shall be made to determine whether preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with table IV. The lot size shall be the number of shipping containers in the inspection lot. The sample unit shall be one shipping container fully packaged. Defects of closure listed in table IV shall be examined on shipping containers fully prepared for delivery.

TABLE IV. Packaging inspection

| Examine | Defects |
|---|---|
| Marking (exterior and interior) | Omitted, incorrect, illegible, or of improper size, location, sequence, or method of application |
| Materials | Any component missing<br>Any component damaged, affecting serviceability |
| Workmanship | Inadequate application of components, such as incomplete closure of rolls, inadequate sewing or sealing of wrapping, improper taping, loose strapping materials, loose ties<br>Bulged or distorted containers |
| Preservation | Preservation improper, improperly applied, or missing |
| Weight | Net weight exceeds requirements |
| Count of sheets | Number of sheets less than marked on container |

4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table VI, shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

the applicable test methods. The sample unit shall be 1 square yard of rolled felt or one sheet of felt, as applicable. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final result. The sample size (number of sample units) shall be as listed in table V.

TABLE V. Sample size for testing

| Lot size in yards or sheets | Sample size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

TABLE VI. Test methods

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Fiber identification | 3.2.1 | 1/ |
| Fiber content | 3.2.1 | 2101 |
| Washed | 3.2.1 | 1/ |
| Extractable matter | 3.2.2 | 2611 2/ |
| Weight | 3.2.3 | 5041 3/ |
| Thickness | 3.2.3 | 5030 |
| Organic acidity | 3.2.7 | 4.4.1 |
| Alkalinity or mineral acidity | 3.2.8 | 2811 |
| Flexibility | 3.3.1 | 4.4.2 4/ |
| Splitting resistance | 3.3.2 | 4.4.3 |
| Bulk density | 3.3.3 | 4.4.4 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

TABLE VI. Test methods - Continued

| |
|---|
| 1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement. |
| 2/ Test method 2611 except that the test specimen shall be cut into pieces of approximately 1/8 inch. |
| 3/ Method 5041 shall be used except that the weight shall be calculated as follows: $\text{Wt/sq yd (pounds)} = \frac{\text{Weight of specimen (grams)} \times 36 \text{ in/yd} \times 36 \text{ in/yd}}{\text{Area of specimens (sq in)} \times 453.59 \text{ (grams/lb)}}$ |
| 4/ One determination per sample unit and the results reported as "pass" or "fail". |

4.4.1 Organic acidity. A specimen totaling 10.0 grams ± 1 gram shall be tested for organic acidity.

4.4.1.1 Procedure. Heat the specimen in an Erlenmeyer flask with 100 milliliters of neutral 95 percent ethyl alcohol at a temperature between 50° and 60°C for 15 minutes. The temperature shall be maintained by means of a water bath. Filter the solution through filter paper, retaining as much of the specimen as possible in the flask. Repeat the process, then wash the filter paper with a small amount of warm alcohol, all filtrates and washing being caught in the same vessel. Cool the solution to room temperature and titrate with standard alcoholic KOH (in range of 0.2N). A blank determination shall be made on all reagents and the necessary correction applied.

4.4.1.2 Calculations. The method of calculating the organic acidity shall be as follows:

a. Highly ionized acids: use methyl orange as indicator. Shall not indicate presence of acid greater than 0.2 percent.

b. Total acidity: use phenolphathalein as indicator.

c. Organic acids (or slightly ionized acids): (b) - (a) = organic (or slightly ionized) acids.



Source: https://assist.dla.mil -- Downloaded: 202610-26T11:04Z
Check the source to verify that this is the current version before use.

d. Acidity as acetic acid percent = $\frac{6.0 \text{ V N}}{W}$

where V = milliliters of standard KOH corrected for blank determination, required to titrate organic acid only,
N = normality of standard KOH, and
W = grams of sample.

e. Two determinations shall be made on each sample unit and the average reported to the nearest 0.1 percent.

4.4.2 Flexibility. All felt shall be bent 160° over a rod equal in diameter to the thickness of the felt. Any visible sign of cracking of the specimen shall be cause for rejection.

4.4.3 Splitting resistance. The felt shall be tested for splitting resistance as follows.

4.4.3.1 Apparatus. The test apparatus shall be as specified in method 5100 except that the width of the jaw shall exceed the specimen width by 0.5 inch and that all machine attachments for determining maximum load shall be disengaged during the test.

4.4.3.2 Procedure. Cut eight specimens with a die 2 inches by 6 inches from each sample unit: four specimens shall be cut in the lengthwise direction and four specimens shall be cut in the widthwise direction. The specimens shall be cut in such a manner that the 6-inch dimension is parallel with the direction of the material being tested. Each specimen shall be evenly split in thickness a distance of 2 inches by cutting with a knife or other sharp instrument. The split shall be within the middle third of the thickness of the specimen. Clamp one of the split-ends in one jaw of the testing machine and the other split-end in the other jaw. Record the maximum load in pounds necessary to split the specimen completely apart to the nearest 0.1 pound. Record the three highest peak-load values as read from the recording dial or autographic recording device. These three separate values from the individual specimen shall be averaged and this average recorded as the single determination of the splitting resistance of the specimen and recorded separately.

4.4.3.3 Report. The splitting resistance of each sample unit shall be obtained by averaging the four single determination values obtained in 4.4.3.2 above for the direction being tested and reported to nearest 0.1 pound.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

4.4.4 Bulk density. Bulk density shall be calculated as follows:

$$\text{Bulk density} = \frac{\text{Weight of felt (lb/sq yd)}}{\text{Thickness of felt (inches)}} \times 0.0214$$

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each felt roll length shall be wound open width on a convolute chipboard tube, except that each felt roll length 1/2 inch or greater in thickness need not be wound on a tube. The ends of the tube shall be flush with or extend not more than 1 inch beyond each side of the width of the rolled felt. A quantity of felt sheets of one description only shall be neatly stacked to form a bundle. Each felt roll or bundle of felt sheets shall be completely wrapped in two thicknesses of 60-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. The paper wrapping shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45. During the rolling or bundling operation, not less than 1 1/2 pounds of naphthalene flakes or chips shall be evenly distributed over each 50 square feet of felt.

5.1.2 Commercial. Felt shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Felt sheets and felt rolls shall be shipped in cloth coverings or packed in shipping containers as specified herein.

5.2.1.1 Cloth-covered shipments. Each felt roll or felt sheets of one description only, preserved as specified in 5.1, shall be completely overwrapped in a waterproof barrier material conforming to type III, grade B or C of MIL-L-10547. The seams and folds of the overwrap barrier material shall be secured by the applicable closure and sealing requirements specified in the appendix of MIL-L-10547. Each overwrapped felt roll or bundle of felt sheets shall then be wrapped and sewn in cotton osnaburg cloth conforming to class 2 of CCC-C-429 or jute (or kenaf) burlap cloth conforming to class 3 of CCC-C-467. The cloth covering shall be securely sewn with types I or II, 16-ply cotton twine conforming to T-T-871 or jute twine conforming to type I, size No. 2 of T-T-911, with approximately one stitch to the inch and every third stitch knotted. Five inches of surplus covering shall be gathered together on each of the four corners and securely sewn into ears for handles.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

Alternatively, the cloth covering may be in bias-sewn tubing form or bags (tubing, presewn on one end) conforming to MIL-T-40625 with the closure effected by securing each open end with two wire ties. The first wire tie shall be applied as close to the roll or bundle as possbile. The second wire tie shall be applied approximately 1 inch from the first wire tie, with the twisted ends positioned opposite to the ends of the first wire tie. The wire ties shall be not less than 6 inches long, 0.072 inch diameter, galvanized soft iron or steel wire with a 1/2-inch diameter formed eye at each end. The weight of contents of each cloth-covered shipment shall not exceed 200 pounds, except when the weight of an individual felt roll exceeds this weight.

5.2.1.2 Shipping containers. Felt rolls or felt sheets of one description only, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC-L, grade V2s or PPP-B-636; class II, style A of PPP-B-591; overseas type, style A or J of PPP-B-601; or class 2, style 2 or 4 of PPP-B-621. Each PPP-B-636 container shall have the inside fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each fiberboard container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Each wood container shall be waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L-10547; and closed and reinforced in accordance with the appendix of the applicable container document. The weight of contents for each PPP-B-636 container shall not exceed 65 pounds. The weight of contents for the other shipping containers shall not exceed 200 pounds, except when the weight of an individual felt roll exceeds this weight.

5.2.2 Level B packing. Felt sheets and felt rolls shall be shipped in cloth coverings or packed in shipping containers as specified herein.

5.2.2.1 Cloth-covered shipments. Each felt roll or felt sheets of one description only, preserved as specified in 5.1, shall be wrapped in a cloth covering as specified in 5.2.1.1. The waterproof barrier material overwrap requirements shall not apply.

5.2.2.2 Shipping containers. Felt rolls or felt sheets of one description only, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636; class 1, style A of PPP-B-591; domestic type style A or J of PPP-B-601; or class 1, style 2 or 4 of PPP-B-621. Each PPP-B-636 container shall have the inside fitted with a box liner conforming to class domestic, variety DW, grade 275 of PPP-B-636. Each fiberboard container shall be closed in accordance with method II of the appendix of PPP-B-636. Each wood container shall be closed and reinforced in accordance with the appendix of the applicable container document. The weight



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

of contents for each PPP-B-636 container shall not exceed 65 pounds. The weight of contents for the other shipping containers shall not exceed 200 pounds, except when the weight of an individual felt roll exceeds this weight.

5.2.2.2.1 Weather-resistant containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

5.2.3 Commercial packing. The felt, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Civil requirements. In addition to any special marking required by the contract or purchase order, unit packs, cloth-covered shipments, and shipping containers shall be marked in accordance with FED-STD-123 or ASTM D 3951, as applicable.

5.3.2 Military requirements. In addition to any special marking required by the contract or purchase order, unit packs, cloth-covered shipments, and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The felt (hair) is intended for uses listed below and for similar applications:

H-26-S and H-26-R - For insulation where the temperature will not exceed 120°F for cushioning, packing, padding, and crating of greater than one-time use. Vibration mountings and dust shields.

H-34-S and H-34-R - For cushioning, packing, padding, and shielding where moderate strength and abrasion resistance are required. Vibration and shock mounts for long-term operation. Ammunition components when class N. Filter elements.

H-43-S and H-43-R - For cushioning, packing, padding, and shielding where high strength and abrasion resistance are required. Vibration and shock mounts for long-term operation and moderate loading. Ammunition components when class N. Polishing blocks, wheels and pads, wiper, applicators, and filter elements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

H-56-S and H-56-R - For cushioning, packing, padding, and shielding where unusual strength, hardness, and abrasion resistance are required. Vibration and shock mounts for heavy loading and high frequency. Ammunition components when class N. Heavy-duty polishing blocks, wheels, and pads. Wipers, applicators, and filters.

6.2 Ordering data. Purchasers should select the preferred option permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type, class, thickness, and weight required (1.2.1).
c. When a first article is required (see 3.1).
d. Length and width or sheet size as applicable, if other than specified (see 3.2.4 and 3.2.5).
e. Color, if other than natural (see 3.2.6).
f. Organic acidity when required (see 3.2.7 and 6.4).
g. Alkalinity and mineral acidity when required (see 3.2.8 and 6.4).
h. When mildew-resistant treatment is required (see 3.2.9).
i. When other than ticket is required (see 3.4.1).
j. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
k. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.2.1).

6.3 First article. When a first article inspection is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for inspection and approval of first article test results, and disposition of first article.

6.4 Organic acidity, alkalinity, and mineral acidity. If felt is to be used in direct contact with brass, bronze, copper, aluminum, or other materials that might be affected by acid, purchasers should specify that the felt shall be free from organic acidity, alkalinity, and mineral acidity (see 3.2.7 and 3.2.8).

6.5 Hair felt types. The hair felt types covered by this specification are classified by a code letter "H" designating the composition of fibers as predominately washed cattle hair. Final letters designate the general form of manufacture, either "R" or "S" for roll or sheet manufacture, respectively, and the relative condition of residual acidity, either "U" for untreated or "N" for neutralized.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians

Army - GL
Navy - NU
Air Force - 99

Review activities

Army - MD
Air Force - 82
DLA - CT

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
VA - OSS
HHS - SSA
USDA - AMS
JUS - FPI

PREPARING ACTIVITY:

Army - GL

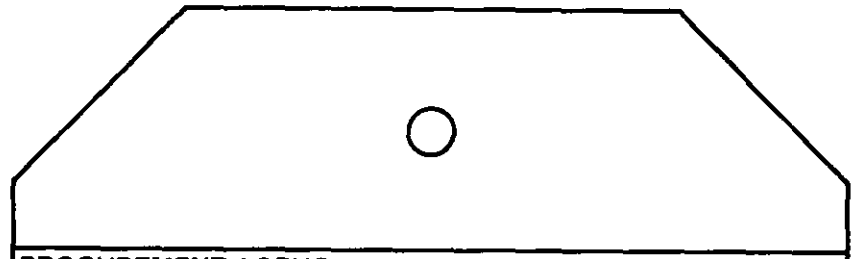
Project Number 8305-0035



Source: https://assist.dla.mil -- Downloaded: 202640-26T11:04Z
Check the source to verify that this is the currentversion before use.

**EXAMPLE IS APPROXIMATE TICKET SIZE DESIRED**

C-F-202F



| | | |
|---|---|---|
| PROCUREMENT AGENCY | | |
| NATIONAL STOCK NUMBER | | |
| NOMENCLATURE | WIDTH | |
| COLOR | | |
| THICKNESS | | |
| CONTRACTOR/SUPPLIER | | |
| FINISHING MILL | | |
| CONTRACT/ORDER NUMBER AND DATE | | |
| SPECIFICATION NUMBER AND DATE | LOT NUMBER | PIECE NUMBER |
| TYPE CLASSIFICATION | | |
| NET YARDS<br>WIDTH | GROSS YARDS<br>LENGTH | |
| INSPECTOR | DATE | |

**NOTE:** ***Bar Code Marking shall be placed on reverse side of the piece ticket.***

Figure 1. Piece Ticket



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. *All users of military standardization documents* are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. *Enter in block 6 any remarks not related to a specific paragraph of the document.* If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE ARMY
US Army Natick Research and Development Center
Natick, Massachusetts 01760

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

**BUSINESS REPLY MAIL**
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Center
ATTN: STRNC-ES
Natick, MA 01760

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:04Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| C-F-202F | Felt Sheet (Hair) and Felt Roll (Hair) |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:04Z
Check the source to verify that this is the current version before use.